

Research Memo - Dominant Transportation Methods

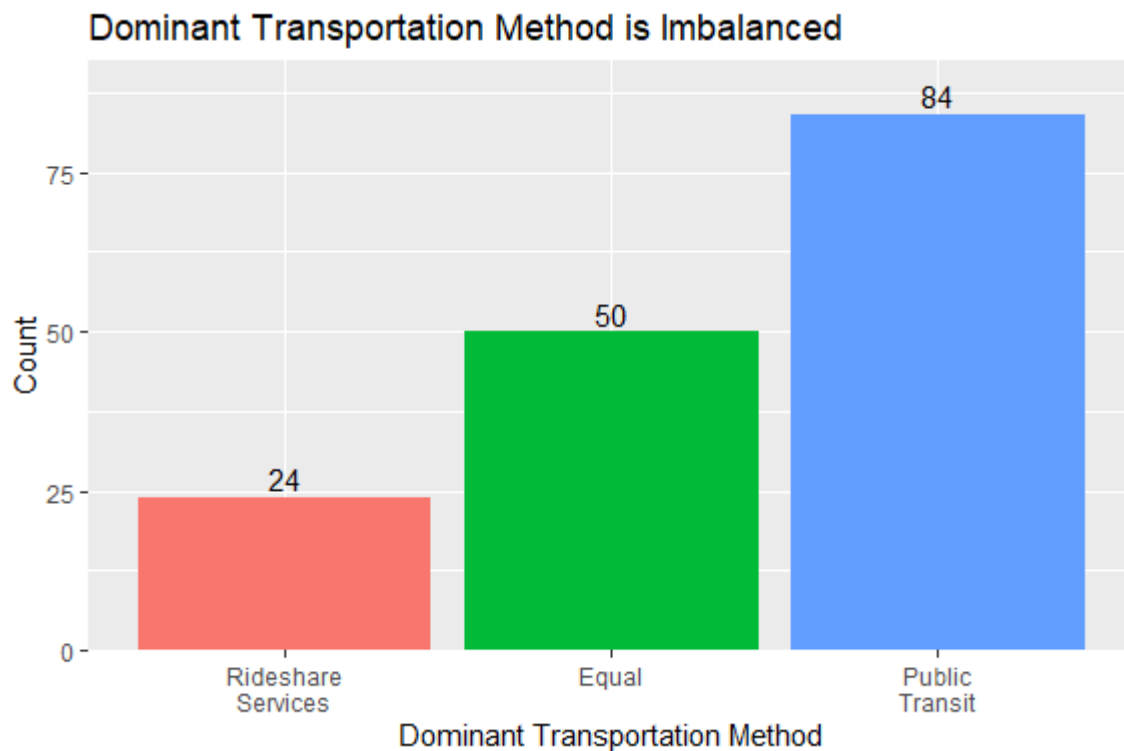
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To: DAGE & Dr. Adam Ross Nelson

Re: [Are there differences in dominant transportation method?]

To determine if our data supports pursuing the open-text classification research question, EDA examined the distribution of respondents' dominant transportation method.



When comparing respondents with a dominating transportation method, public transit is far more prevalent than rideshare services. This large imbalance is problematic for classification and suggest that model training will require substantial resampling.

We also have ~32% of respondents who indicated equal usage across both transportation methods. Pursuing binary classification would entail excluding a significant portion of our sample as a result. While there are still large imbalances between all 3 classes, my recommendation is to pursue a 3-way classification for this problem to avoid reducing our usable training sample.

Next steps include exploration of viable resampling techniques that will assist our models in training and performing well, keeping in mind the complicate nature of our open-text inputs.